

Math 1352 Exam III

Discussion Time and TA:

Name:

Calculators are not allowed on this exam. Work all questions completely. Show all work as described in class. Copyright 2011 Dr. Brock Williams and Texas Tech University. Unauthorized reproduction prohibited.

1. Indicate if the following series converge or diverge. You must identify all the tests you use and show all work needed to apply them.

8 (a) $\sum_{k=0}^{\infty} \frac{5^k}{k!}$

8 (b) $\sum_{k=1}^{\infty} \frac{3 \ln(k)}{k}$

8 (c) $\sum_{k=2}^{\infty} \frac{3^k}{k+4}$

$$\boxed{8} \quad \text{(d)} \quad \sum_{k=0}^{\infty} \frac{5k}{3k^4 + 7}$$

$$\boxed{8} \quad \text{(e)} \quad \sum_{k=0}^{\infty} 5 \left(\frac{2}{3} \right)^k$$

$$\boxed{8} \quad \text{(f)} \quad \sum_{k=1}^{\infty} \frac{1}{k} - \frac{1}{k+1}$$

- 10 2. Does $\sum_{k=1}^{\infty} \frac{(-1)^k}{3\sqrt{k}}$ converge conditionally, converge absolutely, or diverge? Explain completely showing all work for all tests you use.

- 12 3. Find the radius of convergence and interval of convergence of the series $\sum_{k=1}^{\infty} \frac{1}{k} 3^k (x-5)^k$.

10 4. Find the first 3 terms of the Taylor series for $f(x) = \ln x$ about $c = 3$.

8 5. Find the Maclaurin series for $5x^3e^{4x}$.

4 6. Suppose $\sum_{k=0}^{\infty} a_k x^k$ converges when $x = 3$. Give 2 other values of x for which $\sum_{k=0}^{\infty} a_k x^k$ must converge.

8 7. Indicate if the following are always true or may be false.

(a) If $\lim_{k \rightarrow \infty} a_k = 0$, then $\sum a_k$ converges.

(b) If $a_k > b_k \geq 0$ and $\sum b_k$ diverges, then $\sum a_k$ converges.

(c) If $a_k > 0$ and $\lim_{k \rightarrow \infty} \sqrt[k]{a_k} = 0$, then $\sum a_k$ converges.

(d) If $a_k > 0$ and $\lim_{k \rightarrow \infty} \frac{a_k}{1/k} = \frac{1}{2}$, then $\sum a_k$ converges.